Dimitrios S. Nikolopoulos

John W. Hancock Professor of Engineering; IEEE Fellow
Professor, Department of Computer Science
Professor (by courtesy), Department of Electrical and Computer Engineering
Royal Society Wolfson Research Fellow
Fellow: BCS, IET, AAIA, AIIA
Distinguished Member, ACM; Distinguished Visitor, IEEE Computer Society

Virginia Tech
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Employment History

John W. Hancock Professor of Engineering, Virginia Tech	Aug. 2019–present
Professor, Department of Computer Science, Virginia Tech	Aug. 2019–present
Professor (by courtesy), Department of Electrical and Computer Engineering, Virginia Tech	Aug. 2020–present
Associate Director, Stacks@CS Center for Computer Systems Research, Virginia Tech	Aug. 2022–present
Honorary Professor, School of Electronics, Electrical Engineering and Computer Science, Queen's University Belfast	Aug. 2022–present
Director, Queen's University Belfast Research Institute in Electronics, Communications and Information Technology (ECIT)	Jan. 2018–Aug. 2019
Head of School, School of Electronics, Electrical Engineering and Computer Science, Queen's University Belfast	Jan. 2016–Jan. 2018
Royal Society Wolfson Research Fellow	Sep. 2015–present
Professor & Chair in High Performance and Distributed Computing, School of Electronics, Electrical Engineering and Computer Science, Queen's University Belfast	Jan. 2012–Aug. 2019
Director, Center for Data Science and Scalable Computing, Queen's University Belfast	Feb. 2016–Aug. 2019
Director, High Performance and Distributed Computing Research Cluster, Queen's University Belfast	Jan. 2012–Feb. 2016
Adjunct Professor, Department of Computer Science, Virginia Tech	Oct. 2016–Aug. 2019
Adjunct Professor, Department of Computer Science, Old Dominion University	Oct. 2013–present
Associate Professor with Tenure, Department of Computer Science, University of Crete	Sep. 2009–Jan. 2012
Affiliate Professor, Institute of Computer Science, Foundation for Research and Technology (FORTH)	Sep. 2009–Jan. 2012
Associate Professor with Tenure, Department of Computer Science, Virginia Tech	Aug. 2008–Sep. 2009
Associate Professor, Tenure-Track, Department of Computer Science, Virginia Tech	Aug. 2006–Aug. 2008
Assistant Professor, Tenure-Track, Department of Computer Science, College of William & Mary	Aug. 2002–Aug. 2006
Visiting Assistant Professor, Department of Electrical and Computer Engineering, University of Illinois at Urbana–Champaign	Jan. 2001–Aug. 2002

Earned Degrees

Ph.D.	Computer Engineering and Informatics, University of Patras	2000
M.Eng.	Computer Engineering and Informatics, University of Patras	1997
Diploma[1]	Computer Engineering and Informatics, University of Patras	1996

[1] A European Diploma in Engineering is a five-year university degree, academically positioned between a Bachelor's and Master's in the U.S. system.

Primary Research Areas

High-Performance Computing and Computational Science; Computer Systems

Current Technical Research Interests

- **Efficient and resilient AI systems across the computing continuum:** federated learning; edge/microcloud infrastructure; and runtime/orchestration for robustness under heterogeneity and failures.
- **Performance engineering for modern accelerators:** GPU/CPU overlap; memory and KV-cache redesign; long-context inference/training efficiency; and cost/energy-aware optimization.
- **Systems for data-intensive science and cyberinfrastructure:** streaming-plus-learning pipelines; scalable experimentation/digital twins; and deployable research platforms with measurable impact.

Honors and Awards

[43] **Sigma Xi Full Member**, The Scientific Research Honor Society, 2025
[42] **IEEE Fellow**, "For contributions to dynamic execution environments and multiprocessor memory management", 2024
[41] **IEEE Computer Society Distinguished Visitor**, 2024
[40] **Fellow, International Artificial Intelligence Industry Alliance**, 2024
[39] **Fellow, Asia-Pacific Artificial Intelligence Association**, 2023
[38] **Distinguished Contributor, IEEE Computer Society**, 2022
[37] **Best Paper Award**, Design Automation and Test in Europe Conference (DATE), 2020
[36] **IEEE Award for Editorial Excellence**, IEEE Transactions on Parallel and Distributed Systems, 2020
[35] **HiPEAC Paper Award**, twice in 2020 (for ASPLOS 2020 and MICRO 2020 papers)
[34] **Best Paper Award Nomination**, IEEE/ACM International Symposium on Microarchitecture (MICRO), 2020
[33] **John W. Hancock Chair in Engineering**, Virginia Tech, 2019–present
[32] **Elsevier Distinguished Editorial Service Award**, 2019
[31] **Distinguished Member of the ACM**, 2018
[30] **IEEE Outstanding Service Award**, 2018
[29] **Fellow of the IET**, 2017
[28] **Investors in People Silver Award**, 2017
[27] **Royal Society Wolfson Research Merit Award**, 2015 (extended to lifetime)
[26] **SFI-DEL Investigator Award**, 2015
[25] **Fellow of the British Computer Society**, 2014
[24] **IEEE Outstanding Service Award**, 2014

[23] **Best Paper Award**, ACM International Workshop on Code Optimization for Multi and Many Cores (COSMIC), 2013
[22] **ACM Senior Member**, 2011
[21] **IEEE Senior Member**, 2010
[20] **IEEE Outstanding Service Award**, 2010
[19] **Best Paper Award Nomination**, International Conference on Parallel Processing (ICPP), 2010
[18] **Marie Curie Fellow**, 2009
[17] **HiPEAC Fellow**, 2008
[16] **IBM Faculty Award**, 2007
[15] **Best Paper Award**, ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPOPP), 2007
[14] **Best Paper Nomination**, ACM Symposium on High-Performance Parallel and Distributed Computing (HPDC), 2006
[13] **DOE Early Career Principal Investigator Award**, 2005
[12] **Best Paper Award**, International Workshop on OpenMP (IWOMP), 2005
[11] **NSF CAREER Award**, 2004
[10] **Best Paper Award**, International Symposium on High-Performance Computing (ISHPC), 2003
[9] **Best Paper Award**, IEEE/ACM International Symposium on Cluster Computing and the Grid (CCGRID), 2002
[8] **Best Paper Award**, IEEE/ACM International Parallel and Distributed Processing Symposium (IPDPS), 2002
[7] **Best Paper Award Nomination**, ACM International Conference on Supercomputing (ICS), 2001
[6] **Best Paper Award Nomination**, IEEE/ACM Supercomputing: High-Performance Computing, Networking, Storage, and Analysis Conference (SC), 2001
[5] **Best Paper Award**, IEEE/ACM Supercomputing: High-Performance Computing, Networking, Storage and Analysis Conference (SC), 2000
[4] **Best Paper Award Nomination**, ACM International Conference on Supercomputing (ICS), 1999
[3] **Outstanding Academic Performance Award**, Technical Chamber of Greece, 1996
[2] **Outstanding Academic Performance Award**, Technical Chamber of Greece, 1993
[1] **Outstanding Academic Performance Award**, National Scholarships Foundation of Greece, 1992

Publications

Refereed Publications

Journal Articles

[284] X. Li, S. Ghafouri, H. Vandierendonck, D. John, and D. Nikolopoulos, "Sweet: Serving workload-balanced end-to-end efficient and tailored edge inference via quantization and partitioning," ***Frontiers in Complex Systems***, Apr. 2026, Section on Multi- and Cross-Disciplinary Complexity.
[283] X. Li, J. Fan, Q. Wang, D. Spatharakis, S. Ghafouri, H. Vandierendonck, D. John, B. Ji, A. R. Butt, and D. Nikolopoulos, "Wisp: Waste- and interference-suppressed distributed speculative llm serving at the edge via dynamic drafting and slo-aware batching," ***Proceedings of the ACM on Measurement and Analysis of Computing Systems (ACM POMACS-SIGMETRICS)***, 2026.

[282] A. Kumar, C. O'Mahoney, P. K. Werle, S. Shanker, D. Nikolopoulos, B. Ji, H. Vandierendonck, and D. John, "Marvel: An end-to-end framework for generating model-class aware custom risc-v extensions for lightweight ai," ***IEEE Open Journal of Circuits and Systems***, vol. 6, pp. 445–456, 2025.

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[279] C.-H. Hong, M. Lee, H. Ahn, and D. Nikolopoulos, "Gshare: A centralized gpu memory management framework to enable gpu memory sharing for containers," ***Future Generation Computer Systems***, vol. 130, pp. 181–192, May 2022.

[278] K. Dichev, D. di Sensi, I. Spence, K. Cameron, and D. Nikolopoulos, "Power log'n'roll: Power-efficient localized rollback for mpi applications using logging protocols," ***IEEE Transactions on Parallel and Distributed Systems***, vol. 33, no. 6, pp. 1276–1288, Jun. 2022.

[277] U. Minhas, R. Woods, D. Nikolopoulos, and G. Karakonstantis, "Efficient, dynamic multi-task execution on fpga-based computing systems," ***IEEE Transactions on Parallel and Distributed Systems***, vol. 33, no. 3, pp. 710–722, Mar. 2022.

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[275] J. Lee, H. Vandierendonck, and D. Nikolopoulos, "Mixed precision kernel recursive least squares," ***IEEE Transactions on Neural Networks and Learning Systems***, no. 3, pp. 1284–1298, Mar. 2022. DOI: 10.1109/TNNLS.2020.3041677.

[274] J. Sun, H. Vandierendonck, and D. Nikolopoulos, "Fast load balance parallel graph analytics with an automatic data structure selection algorithm," ***Future Generation Computer Systems***, vol. 112, pp. 612–623, Nov. 2020.

[273] J. Lee, G. Peterson, H. Vandierendonck, and D. Nikolopoulos, "Air: Iterative refinement acceleration using arbitrary dynamic precision," ***Parallel Computing***, vol. 97, Sep. 2020, Article: 102663.

[272] N. Wang, B. Varghese, M. Matthaiou, and D. Nikolopoulos, "Dyverse: Dynamic vertical scaling in multi-tenant edge environments," ***Future Generation Computer Systems***, vol. 108, pp. 598–612, Jul. 2020.

[271] H. Vandierendonck and D. Nikolopoulos, "Hyperqueues: Design and Implementation of Deterministic Concurrent Queues," ***ACM Transactions on Parallel Computing***, vol. 6, no. 4, pp. 1–35, Nov. 2019, Article 23.

[270] A. Hassan, H. Vandierendonck, and D. Nikolopoulos, "Fast and Energy-Efficient OLAP Data Management on Hybrid Main Memory Systems," ***IEEE Transactions on Computers***, vol. 63, no. 11, pp. 1597–1611, Nov. 2019.

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[244] P. Gschwandtner, C. Chalios, D. S. Nikolopoulos, H. Vandierendonck, and T. Fahringer, "On the Potential of Significance-Driven Execution for Energy-Aware HPC," ***Computer Science – Research and Development***, vol. 30, no. 2, pp. 197–206, Feb. 2015.

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[240] H. Vandierendonck, G. Tzenakis, and D. Nikolopoulos, "Analysis of Dependence Tracking Algorithms for Task Dataflow Execution," ***ACM Transactions on Architecture and Code Optimization***, vol. 10, no. 4, pp. 1–24, Dec. 2013, Article No. 61.

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threaded Architectures," ***Journal of Parallel and Distributed Computing***, vol. 69, no. 7, pp. 601–612, Jul. 2009.

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Papers in Archival Conference Proceedings

[215] F. Amin, S. Afroz, K. Gharami, M. Moghadampanah, and D. Nikolopoulos, "Diffpro: Joint timestep and layer-wise precision optimization for efficient diffusion inference," in *Proceedings of the 19th European Conference on Computer Vision* ***(ECCV)***, Malmö, Sweden, Sep. 2026.

[214] Y. Shen and D. Nikolopoulos, "Stride-aware page prefetching for gpu unified memory via markov pattern prediction," in *Proceedings of the 2026 ACM SIGPLAN International Symposium on Memory Management* ***(ISMM)***, Boulder, CO, Jun. 2026.

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Theses

[2] D. Nikolopoulos, "System Software Support for Reducing Memory Latency on CC-NUMA Architectures," PhD Dissertation, Department of Computer Engineering and Informatics, University of Patras, Dec. 2000.

[1] D. Nikolopoulos and I. Tsolakis, "Load Balancing in Volunteer Computing Clusters," MSc Thesis, Department of Computer Engineering and Informatics, University of Patras, Jul. 1997.

Publication Metrics

Citation metrics (last verified May 2026)

Source	Citations	h-index	Other
Google Scholar	7,942	43	i-10 index: 148
Scopus	4,192	31	FWCI: 1.39

Top 2% of Computer Scientists Worldwide according to Scopus Standardized Citation Indicators.

Knowledge and Technology Transfer & Commercialization

Patents

Research on memory management for stream processing and in-memory database systems, supported by NSF grants 0715051, 0720673, 0904844, led to foundational results later extended into U.S. Patents (1) 9,112,666 by former student Scott Schneider, and (2) 10,365,997, (3) 10,387,127, (4) 10,474,557, (5) 10,698,732 by former student Ahmad Hassan. In accordance with NSF and other federal funding agency policies on open dissemination, I declined to participate in inventorship but contributed to the original publication. Patent details below:

[1] "Elastic auto-parallelization for stream processing applications based on measured throughput and congestion," U.S. Patent 9,112,666 (granted Aug. 18, 2015).

[2] "Optimizing DRAM memory based on read-to-write ratio of memory access latency," U.S. Patent 10,365,997 (granted Jul. 30, 2019).

[3] "Detecting sequential access data and random access data for placement on hybrid main memory for in-memory databases," U.S. Patent 10,387,127 (granted Aug. 20, 2019).

[4] "Source code profiling for line-level latency and energy consumption estimation," U.S. Patent 10,474,557 (granted Nov. 12, 2019).

[5] "Page ranking in operating system virtual pages in hybrid memory systems," U.S. Patent 10,698,732 (granted Jun. 30, 2020).

Technology Transfer Activities

- Train-less deep neural network architecture search (**IBM Watson Studio**, 2019)
- Query planning and data placement for in-memory transactional–analytical databases (**SAP**, 2019)
- High-level program parallelization software for FPGAs (**Crevinn**, 2019)
- Automatic scaling of streaming operators (**IBM**, 2010)
- Adoption of methods for memory management and dynamic software scaling (**Samsung, Red Hat, Microsoft, Oracle, IBM, Alpine Electronics, Thales, Reservoir Labs**, 2008–present)

Research Grants

Funding summary (as of July 26, 2026)

Total grant value across 54 awards: **$141.8M**

PI share (budget under my direct PI/Co-PI leadership): **$34.5M**

Currency conversion at year-of-award annual average rates. Major sponsors: NSF, DOE, EU Horizon 2020 & FP7, EPSRC, Royal Society, Marie Curie, IBM, Intel, SAP, Sony, Cisco. Totals exclude in-kind allocations unless noted.

Upcoming / awarded

[54] **[08/2026–07/2027] Eliminating Redundant Computation in Multi-Agent LLM Systems via Hybrid KV-Cache Communication**.
Amazon. PI. $75,000 total (PI share: $37,500).

[53] **[05/2026–05/2027] SCALE-R: Semantic Co-rendering & Adaptive Link-aware Encoding for Real-time Graphics**.
Sony. PI. $150,000 total (PI share: $75,000).

Active and completed

[52] **[10/2025–10/2026] Cloud–Edge–Neuromorphic Computing Integration for Multimodal Sensing and AI-Driven Modeling of Complex Biological Systems**.
Virginia Tech Institute for Critical Technology & Applied Science / Institute for Advanced Computing. Co-PI. $50,000 total (PI share: $12,500).

[51] **[08/2025–08/2026] Advancing AI-Driven Autonomous Experimentation for Turbulent Flows**.
Virginia Tech College of Engineering Major Grant Initiative Program. PI. $50,000.

[50] **[05/2024–11/2024] GPU-Accelerated High-Performance Computing to Supercharge AI Models for Pandemic Prevention**.
National Science Foundation (National AI Resource). Co-PI. In-kind HPC resource allocation (no direct $ amount).

[49] **[10/2023–09/2026] SWEET: Hardware and Software for Sustainable Wearable Edge Intelligence**.
National Science Foundation (#2315851). PI. $600,000.

[48] **[08/2023–07/2024] WASML: Enabling Scalable Serverless Computer Vision**.
Sony Faculty Innovation Award (#VTF-AG3ZURVF). PI. $100,000.

[47] **[02/2022–01/2023] VirtEdgeAI: Seamless Virtualization of AI Accelerators in Edge Computing Substrates**.
Cisco (#VTF-446663). PI. $100,000.

[46] **[06/2021–05/2024] Collaborative Research: CNS Core: Medium: HardLambda: A new FaaS Abstraction for Cross-Stack Resource Management in Disaggregated Datacenters**.
National Science Foundation (#2106634). Co-PI. $600,000 (PI share: $210,000).

[45] **[12/2017–12/2018] Belfast Region City Deal, Queen's Global Innovation Institute**.
UK Treasury (77%); Belfast City Council (23%). PI. £56,000,000 (PI share: £15,000,000).

[44] **[01/2018–06/2021] Biohaviour: Building the Blind Watchmaker**.
EPSRC (#EP/R003564/1). Co-PI. £778,226 (PI share: £194,556).

[43] **[08/2017–07/2018] Variability in Computing Systems**.
Royal Academy of Engineering, Distinguished Visiting Fellowships. Host PI. £4,100.

[42] **[01/2017–12/2020] Scalable, Virtualized Data Center Acceleration**.
Intel. Co-PI. £3,945 (PI share: £795).

[41] **[01/2017–12/2020] OPRECOMP: Open Transprecision Computing**.
EU, Horizon 2020 (#H2020-732631). PI. €5,999,510 (£5,091,933) (PI share: €705,625 (£599,781)).

[40] **[02/2016–01/2019] UNISERVER: A Universal Micro-server Ecosystem by Exceeding the Energy and Performance Scaling Boundaries**.
EU, Horizon 2020 (#H2020-688540). Co-PI. €4,815,810 (£4,333,717) (PI share: €322,648 (£222,516)).

[39] **[02/2016–01/2019] VINEYARD: Versatile Integrated Accelerator-based Heterogeneous Datacenters**.
EU, Horizon 2020 (#H2020-687628). PI. €6,283,895 (£4,467,972) (PI share: €663,625 (£471,850)).

[38] **[10/2015–09/2020] Principles and Practice of Near Data Computing**.
Royal Society Wolfson Research Merit Award (#WM150009). PI. £50,000.

[37] **[09/2015–09/2018] Meeting the Future Challenges of Heterogeneous and Extreme Scale Parallel Computing**.
SFI-DEL, Investigator Awards (#14/IA/2474). PI. £521,947.

[36] **[10/2015–10/2018] ECOSCALE: Energy Efficient Heterogeneous Computing at Exascale**.
EU, Horizon 2020 (#H2020-671632). PI. €4,237,398 (£2,922,346) (PI share: €696,750 (£480,518)).

[35] **[10/2015–10/2018] ALLScale: An Exascale Programming, Multi-objective Optimization and Resilience Management Environment Based on Nested Recursive Parallelism**.
EU, Horizon 2020 (#H2020-671603). PI. €3,366,196 (£2,463,217) (PI share: €438,578 (£320,930)).

[34] **[08/2015–09/2017] Crevinn Teoranta OpenCL Knowledge Transfer**.
Knowledge Transfer Partnership / Intertrade Ireland Fusion Project. PI. £39,000.

[33] **[03/2015–03/2018] SERT: Scale-Free, Energy-Efficient and Resilient CSE Software for Mega-Core Systems**.
EPSRC (Software for the Future II) (#EP/M01147X/1). PI. £963,929 (PI share: £694,909).

[32] **[01/2015–01/2018] RAPID: Heterogeneous Secure Multi-level Remote Acceleration Service for Low-Power Integrated Systems and Devices**.
EU, Horizon 2020 (#H2020-644312). PI. €2,203,800 (£1,695,231) (PI share: €326,925 (£251,481)).

[31] **[01/2015–01/2018] DIVIDEND: Distributed Heterogeneous Vertically Integrated Energy Efficient Data Centers**.
EPSRC, CHIST-ERA (#EP/M015742/1). PI. €1,346,885 (£1,077,508) (PI share: €279,646 (£223,717)).

[30] **[10/2014–10/2017] HPDCJ: Heterogeneous Parallel and Distributed Computing with Java**.
EPSRC, CHIST-ERA (#EP/M015750/1). PI. €1,721,010 (£1,376,808) (PI share: €178,159 (£142,527)).

[29] **[03/2014–03/2017] ASAP: An Adaptive, Highly Scalable Analytics Platform**.
European Commission, FP7-ICT (#FP7-619706). Co-PI. €2,245,128 (£1,909,122) (PI share: €407,720 (£346,701)).

[28] **[01/2014–02/2014] US-Ireland R&D Partnership Planning Grant: Cloud-based Electronic Integration of Patient Records (CLEAR)**.
(#PG20). PI. £1,356.

[27] **[10/2013–10/2016] ENPOWER: Energy-Proportional Computing with Heterogeneous and Reconfigurable Processors**.
EPSRC (#EP/L004232/1). PI. £741,043 (PI share: £348,325).

[26] **[09/2013–09/2016] ALEA: Abstraction-Level Energy Accounting and Optimization in Many-Core Programming Languages**.
EPSRC, System Approaches to Distributed and Embedded Architectures (#EP/L000555/1). PI. £669,561 (PI share: £394,025).

[25] **[09/2013–09/2016] NanoStreams: A Hardware and Software Stack for Real-Time Analytics on Fast Data Streams**.
European Commission, FP7-ICT, Objective 3 (#FP7-610509). PI. €3,300,000 (PI share: €723,565).

[24] **[10/2013–10/2016] CACTOS: Context-Aware Cloud Topology Optimization and Simulation**.
European Commission, FP7-ICT, Objective 1 (#FP7-610811). PI. €3,215,751 (PI share: €583,330).

[23] **[08/2013–08/2019] SAP: PhD Project on High Availability for Petascale Systems**.
SAP AG (#UK-2013-009). PI. £12,436.

[22] **[06/2013–06/2016] SCORPIO: Significance-Based Computing for Reliability and Power Optimization**.
European Commission, FP7-FET-Open (#FP7-323872). PI. €1,890,775 (PI share: €273,400).

[21] **[04/2013–04/2015] NovoSoft: Software Management of Non-Volatile Memory Hierarchies**.
European Commission, Marie Curie Intra-European Fellowship (#FP7-327744). Scientist in Charge. €309,235.

[20] **[03/2013–03/2016] Characterizing and Optimizing In-Memory Database Systems for Emerging Memory Technologies**.
SAP UK Limited (#R502). Co-PI. £34,298 (PI share: £17,149).

[19] **[09/2012–09/2015] GEMSCLAIM: Greener Mobile Systems by Cross Layer Integrated Energy Management**.
EPSRC, CHIST-ERA (#EP/K017594/1). PI. €1,776,688 (PI share: €436,884).

[18] **[11/2012–06/2013] Exascale Mesh Generation Runtime Systems**.
Royal Academy of Engineering, Distinguished Visiting Fellowships. Host PI. £4,100.

[17] **[01/2012–01/2016] HOLISTIC: Hardware and Software Techniques for Multicore Processor Architectures Reliability Enhancement**.
Greek Ministry of Education, Lifelong Learning and Religious Affairs, Thales Programme (#1103). PI. €600,000 (PI share: €98,000).

[16] **[03/2010–03/2012] SCC–MR: Scalable and Energy-Efficient Runtime Support for the MapReduce Programming Model on the Intel SCC**.
Intel Corporation. PI. Equipment donation.

[15] **[06/2010–09/2012] TEXT: Towards Exascale Applications**.
European Commission, FP7-INFRASTRUCTURES Programme (#ICT-261580). PI. €2,470,000 (PI share: €299,364).

[14] **[06/2010–06/2011] ReMap: Rearchitecting MapReduce for Multicore Systems with Explicit Communication**.
High Performance and Embedded Architectures and Compilers Network of Excellence, Cluster Collaboration Grant (#ICT-217068). PI. €3,000 share.

[13] **[03/2010–03/2013] ENCORE: Enabling Technologies for a Programmable Many-core**.
European Commission, FP7-ICT, Objective 3 (#ICT-248647). Co-PI. €2,533,000 (PI share: €533,000).

[12] **[08/2009–08/2013] Coupled Models of Diffusion and Individual Behavior over Extremely Large Scale Social Networks**.
National Science Foundation, PetaApps Program (#OCI-0904844). Co-PI. $1,182,798 (PI share: $200,000).

[11] **[01/2009–01/2013] I-Cores: Hypervisor-based Synthesis of Custom Execution Environments**

for Multi-core Systems.
European Commission, FP7 Programme, Marie Curie International Reintegration Grants (#IRG-224759). PI. €100,000.

[10] **[01/2008–02/2008] HiPEAC Fellowship: Runtime Systems for Parallel Programming**.
European Commission, FP7 Programme, European Network of Excellence in High Performance and Embedded Architectures (#ICT-217068). PI. €8,600 share.

[9] **[07/2007–07/2010] VT-ASOS: Virtualization Technologies for Application-Specific Operating Systems on Many-Core HPC Systems**.
NSF Computer Systems Research Program (#CNS-0720673). PI. $300,000 (PI share: $150,000).

[8] **[07/2007–07/2010] Thermal Conductors: Runtime Software Support for Proactive Heat Management in Advanced Execution Systems**.
NSF Computer Systems Research Program (#CNS-0720750). Co-PI. $350,000 (PI share: $175,000).

[7] **[05/2007–05/2008] Models and Adaptive Runtime Systems for Accessible Parallel Programming on IBM Multi-Core Systems**.
IBM Faculty Award Program (#VTF-874197). PI. $15,000.

[6] **[07/2007–07/2008] MISER: A High-Performance, Power-Aware Cluster**.
NSF Computing Research Infrastructure Program (#CNS-0709025). Co-PI. $500,000 (PI share: $166,667).

[5] **[08/2006–08/2007] Faculty Startup Grant**.
Virginia Tech. PI. $100,000.

[4] **[08/2005–08/2008] MELISSES: Liquid Services for Scalable Multithreaded and Multicore Execution on Emerging Supercomputers**.
DOE Early Career Principal Investigator Award Program (#DE-FG02-06ER25751, DE-FG02-05ER25689). PI. $299,907.

[3] **[05/2005–05/2008] Acquisition of STEMS: A Laboratory for End-to-End Development of Software and Tools for Emerging Multigrain Supercomputers**.
NSF Major Research Instrumentation Program (#CNS-0521381). PI. $228,134 (PI share: $76,045).

[2] **[05/2005–08/2005] A Unified Framework for Multilevel Parallelization on Deep Computing Systems**.
NSF Research Experiences for Undergraduates Program (#CCF-0531887). PI. $6,000.

[1] **[01/2004–01/2009] A Unified Framework for Multilevel Parallelization on Deep Computing Systems**.
NSF CAREER Award Program (#CCF-0346867, CCF-0715051). PI. $419,835 (PI share: $225,000).

Education

Courses Taught

[43] **[Spring 2026]** CS2506: Computer Organization II, Virginia Tech (Teaching Effectiveness: N/A; Class size: 200)

[42] **[Spring 2025]** CS5914: Special Topics in Computer Science: LLMs for High-Performance Code Generation, Virginia Tech (Teaching Effectiveness: 5.70/6.00; Class size: 20)

[41] **[Fall 2024]** CS2506: Computer Organization II, Virginia Tech (Teaching Effectiveness: 4.90/6.00; Class size: 195)

[40] **[Spring 2024]** CS2506: Computer Organization II, Virginia Tech (Teaching Effectiveness: 5.10/6.00; Class size: 150)

[39] **[Fall 2023]** CS5510/ECE5510: Multiprocessor Programming, Virginia Tech (Teaching Effectiveness: N/A; Class size: 45)

[38] **[Spring 2023]** CS2506: Computer Organization II, Virginia Tech (Teaching Effectiveness: 4.87/6.00;

Class size: 192)
[37] **[Fall 2022]** CS5914: Warehouse Scale Computing, Virginia Tech (Teaching Effectiveness: 5.79/6.00; Class size: 17)
[36] **[Spring 2022]** CS2506: Computer Organization II, Virginia Tech (Teaching Effectiveness: 5.16/6.00; Class size: 216)
[35] **[Fall 2021]** CS5510/ECE5510: Multiprocessor Programming, Virginia Tech (Teaching Effectiveness: 5.78/6.00; Class size: 35)
[34] **[Spring 2021]** CS2506: Computer Organization II, Virginia Tech (Teaching Effectiveness: 5.15/6.00; Class size: 148)
[33] **[Fall 2020]** CS4284: Systems & Networking Capstone, Virginia Tech (Teaching Effectiveness: 5.85/6.00; Class size: 23)
[32] **[Spring 2020]** CS2506: Computer Organization II, Virginia Tech (Teaching Effectiveness: 5.16/6.00; Class size: 82)
[31] **[Spring 2015]** ECS1002: Design Projects, Queen's University Belfast
[30] **[Fall 2015]** ECS2001: Second Stage Design Projects, Queen's University Belfast
[29] **[Spring 2014]** ECS1002: Design Projects, Queen's University Belfast
[28] **[Fall 2013]** ECS2001: Second Stage Design Projects, Queen's University Belfast
[27] **[Spring 2013]** ECS1002: Design Projects, Queen's University Belfast
[26] **[Fall 2012]** ECS2001: Second Stage Design Projects, Queen's University Belfast
[25] **[Fall 2011]** CS425: Computer Systems Architecture, University of Crete
[24] **[Fall 2011]** CS100: Introduction to Computer Science, University of Crete
[23] **[Spring 2011]** CS225: Computer Organization, University of Crete
[22] **[Spring 2011]** CS529: Multi-core Systems Programming, University of Crete
[21] **[Fall 2010]** CS425: Computer Systems Architecture, University of Crete
[20] **[Spring 2010]** CS225: Computer Organization, University of Crete
[19] **[Spring 2010]** CS529: Multi-core Systems Programming, University of Crete
[18] **[Fall 2009]** CS425: Computer Systems Architecture, University of Crete
[17] **[Spring 2009]** CS529: Multi-core Systems Programming, University of Crete
[16] **[Spring 2009]** CS225: Computer Organization, University of Crete
[15] **[Fall 2008]** CS425: Computer Systems Architecture, University of Crete
[14] **[Spring 2008]** CS425: Computer Systems Architecture, University of Crete
[13] **[Fall 2007]** CS5234: Advanced Parallel Computation, Virginia Tech
[12] **[Fall 2007]** CS2504: Introduction to Computer Organization, Virginia Tech
[11] **[Spring 2007]** CS2504: Introduction to Computer Organization, Virginia Tech
[10] **[Fall 2006]** CS4234: Parallel Computation, Virginia Tech
[9] **[Spring 2006]** CSCI644: Advanced Computer Architecture, College of William and Mary
[8] **[Fall 2005]** CSCI444/544: Principles of Operating Systems, College of William and Mary
[7] **[Spring 2005]** CSCI644: Advanced Computer Architecture, College of William and Mary
[6] **[Fall 2004]** CSCI444/544: Principles of Operating Systems, College of William and Mary
[5] **[Spring 2004]** CSCI644: Advanced Computer Architecture, College of William and Mary
[4] **[Fall 2003]** CSCI444/544: Principles of Operating Systems, College of William and Mary
[3] **[Spring 2003]** CSCI644: Advanced Computer Architecture, College of William and Mary
[2] **[Fall 2002]** CSCI444/544: Principles of Operating Systems, College of William and Mary
[1] **[Spring 2002]** ECE291: Computer Engineering II, University of Illinois at Urbana–Champaign

Seminars Taught

- **[Spring 2012]** Implementation of Multi-core Programming Models, Universitat Politècnica de Catalunya

- **[Spring 2010]** Multi-core Systems Programming and Optimization, Universitat Politècnica de Catalunya
- **[Spring 2008]** Multi-core Systems Programming and Optimization, Universitat Politècnica de Catalunya
- **[Spring 2007]** Multi-core Systems Programming and Optimization, Universitat Politècnica de Catalunya
- **[Spring 2004]** Multithreaded Architectures and Software, College of William and Mary

Course and Curriculum Development

- CS2506: Computer Organization II — redeveloped to introduce a new open-source processor and tools (Virginia Tech, 2022)
- ECS2001: Software and Electronic Systems Engineering Design Projects (2nd stage) — developed from scratch (Queen's University Belfast, 2016)
- ECS1002: Software and Electronic Systems Engineering Design Projects (1st stage) — developed from scratch (Queen's University Belfast, 2016)
- CS529: Multi-core Processor Programming — developed from scratch (University of Crete, 2009)
- CS425: Computer Systems Architecture — major revision (University of Crete, 2009)
- CS225: Computer Organization — major revision (multi-core, cache coherence) (University of Crete, 2009)
- CS5234: Advanced Parallel Computation — developed from scratch (Virginia Tech, 2006)

Teaching Grants

[4] **[05/2012–06/2012] Advanced Topics in Implementing Multicore Programming Models**. Universitat Politècnica de Catalunya. PI. €2,400.
[3] **[05/2010–06/2010] Multi-core Systems Programming**. Universitat Politècnica de Catalunya. PI. €2,400.
[2] **[05/2008–06/2008] Multi-core Systems Programming**. Universitat Politècnica de Catalunya. PI. €3,600.
[1] **[05/2007–06/2007] Multi-core Systems Programming and Optimization**. Universitat Politècnica de Catalunya. PI. €3,600.

Supervision

Ph.D. Thesis Students

Primary Advisor

[13] *Current Ph.D. students (primary advisor): 7*.
[12] **Dr. Emadeldin Abdrabou** — Electrical and Computer Engineering, Virginia Tech (Dec. 2025). Thesis title: *On the optimization of edge server scaling and placement for Open Radio Access Network (O-RAN) slicing*. (Co-advised with Scott Midkiff.)
[11] **Dr. Esha Barlaskar** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Nov. 2020). Thesis title: *User-centric cloud application management*. (Co-supervised with Ivor Spence and Peter Kilpatrick.)

[10] **Dr. Nan Wang** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Sep. 2019). Thesis title: *Resource management for edge computing systems*.

[9] **Dr. Roxana Istrate** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Jul. 2019). Thesis title: *Efficient neural network architecture search*.

[8] **Dr. Sakil Barbhuiya** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Jul. 2018). Thesis title: *Anomaly detection in cloud and mobile devices*. (Co-supervised with Peter Kilpatrick.)

[7] **Dr. Charalambos Chalios** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Jul. 2017). Thesis title: *Software-defined significance-based computing*. (Co-supervised with Hans Vandierendonck.)

[6] **Dr. Giorgis Georgakoudis** — Computer and Telecommunications Engineering, University of Thessaly (May 2016). Thesis title: *Scheduling and performance characterization on heterogeneous computing systems*. (Co-supervised with Spyros Lalis.)

[5] **Dr. Jae-seung Yeom** — Computer Science, Virginia Tech (May 2014). Thesis title: *Optimizing data accesses for scaling data-intensive scientific applications*. (Co-supervised with Madhav Marathe.)

[4] **Dr. Spyros Lyberis** — Computer Science, University of Crete (Jul. 2013). Thesis title: *Myrmics: A scalable runtime system for global address spaces*.

[3] **Dr. Scott Schneider** — Computer Science, Virginia Tech (Dec. 2010). Thesis title: *Shared memory abstractions for heterogeneous multicore processors*.

[2] **Dr. Filip Blagojevic** — Computer Science, Virginia Tech (May 2008). Thesis title: *Scheduling on asymmetric parallel architectures*.

[1] **Dr. Matthew Curtis-Maury** — Computer Science, Virginia Tech (Mar. 2008). Thesis title: *Improving the efficiency of parallel applications on multithreaded and multicore systems*. **Virginia Tech Outstanding Ph.D. Dissertation Award.**

Co-Advisor

[16] *Current Ph.D. students (co-advisor): 2*.

[15] **Dr. Yuze Li** — Computer Science, Virginia Tech (May 2026). Thesis title: *Cross-stack improvement on memory efficiency*. (Co-supervised with Ali R. Butt.)

[14] **Dr. Abdullahi Abubakar** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Dec. 2023). Thesis title: *Anomaly detection in longitudinal data with applications in cloud computing and healthcare*.

[13] **Dr. Ioannis Tsiokanos** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Dec. 2021). Thesis title: *Cross-layer instruction-aware timing error mitigation and evaluation for energy-efficient dependable architectures*. (Co-supervised with Georgios Karakonstantis.)

[12] **Dr. Konstantinos Tovletoglou** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Jun. 2021). Thesis title: *Modeling and design of energy-efficient dependable memory subsystems*. (Co-supervised with Georgios Karakonstantis.)

[11] **Dr. Jiawen Sun** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Jul. 2017). Thesis title: *The GraphGrind framework: fast graph analytics on large shared-memory systems*. (Co-supervised with Hans Vandierendonck.)

[10] **Dr. Stuart McCool** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Jul. 2016). Thesis title: *Guidance environments for program parallelization and analysis*. (Co-supervised with Peter Kilpatrick.)

[9] **Dr. Ahmad Hassan** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Jul. 2016). Thesis title: *Software management of hybrid main memory systems*.

(Co-supervised with Hans Vandierendonck.)

[8] **Dr. Eoghan O'Neill** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Oct. 2015). Thesis title: *A framework for managing shared accelerators in heterogeneous environments*. (Co-supervised with Peter Kilpatrick.)

[7] **Dr. Aleksandr Khasymski** — Computer Science, Virginia Tech (Feb. 2015). Thesis title: *Accelerated storage systems*. (Co-supervised with Ali R. Butt.)

[6] **Dr. Chun-Yi Su** — Computer Science, Virginia Tech (Dec. 2014). Thesis title: *Energy-aware thread and data management in heterogeneous multi-core and multi-memory systems*. (Co-supervised with Kirk W. Cameron.)

[5] **Dr. Vassilis Papaefstathiou** — Computer Science, University of Crete (Nov. 2013). Thesis title: *Architectural support for software-guided energy reduction of manycore communication*. (Co-supervised with Manolis Katevenis.)

[4] **Dr. Muhammad Mustafa Rafique** — Computer Science, Virginia Tech (Sep. 2011). Thesis title: *An adaptive framework for managing heterogeneous many-core clusters*. (Co-supervised with Ali R. Butt.)

[3] **Dr. Dong Li** — Computer Science, Virginia Tech (Jan. 2011). Thesis title: *Scalable and energy-efficient execution methods for multicore systems*. (Co-supervised with Kirk W. Cameron.)

[2] **Dr. John Christian Linford** — Computer Science, Virginia Tech (May 2010). Thesis title: *Accelerating atmospheric modeling through emerging multi-core technologies*. (Co-supervised with Adrian Sandu.)

[1] **Dr. Richard Tran Mills** — Computer Science, College of William & Mary (Nov. 2004). Thesis title: *Dynamic adaptation to CPU and memory load in scientific applications*. (Co-supervised with Andreas Stathopoulos.)

Postdoctoral Research Fellows

[17] **Dr. Dimitris Spatharakis** — Visiting postdoctoral fellow, School of Electrical and Computer Engineering, National Technical University of Athens (04/2025–05/2025).

[16] **Dr. Sakil Barbhuiya** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (05/2018–08/2019). Research themes: Cloud-based distributed manufacturing; smart manufacturing as a service.

[15] **Dr. Jun-Kyu Lee** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (04/2017–12/2020). Research themes: Approximate computing.

[14] **Dr. Damon Fenacci** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (01/2017–01/2018). Research themes: Memory management.

[13] **Dr. Giorgis Georgakoudis** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (05/2016–09/2018). Research themes: System software and hardware/software interface.

[12] **Dr. Blesson Varghese** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (01/2016–06/2017). Research themes: Energy-efficient and resilient high-performance computing.

[11] **Dr. Kiril Dichev** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (10/2015–08/2019). Research themes: Exascale resilience.

[10] **Dr. Cheol-Ho Hong** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (06/2015–12/2017). Research themes: Accelerator virtualization.

[9] **Dr. Lev Mukhanov** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (04/2014–12/2020). Research themes: Abstraction-level energy accounting in many-core programming languages.

[8] **Dr. Zafeirios Papazachos** — Electronics, Electrical Engineering and Computer Science, Queen's

University Belfast (01/2014–08/2019). Research themes: Data center performance and reliability monitoring and optimization.

[7] **Dr. Hemant Mehta** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (10/2015–04/2017). Research themes: Virtual machine energy accounting.

[6] **Dr. Yun Wu** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (01/2014–01/2017). Research themes: Energy-proportional many-core computing systems.

[5] **Dr. Paul Harvey** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (01/2016–09/2016). Research themes: Exascale programming models.

[4] **Dr. Ahmed Sayed** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (09/2014–02/2015). Research themes: System software for real-time in-memory analytics.

[3] **Dr. Konstantina Mitropoulou** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (01/2014–03/2014). Research themes: System software for real-time in-memory analytics.

[2] **Dr. Hans Vandierendonck** — Institute of Computer Science, Foundation for Research and Technology (FORTH) (10/2010–10/2011). Research themes: Parallel programming; scheduling.

[1] **Dr. Christos D. Antonopoulos** — Computer Science, College of William & Mary (06/2004–06/2006). Research themes: Energy-efficient parallel computation; runtime systems; memory management.

Pre-Doctoral Research Assistants

[4] **Kai Chen** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (in progress). Research area: *Multi-scale power and performance modeling*.

[3] **George Tzenakis** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (10/2012–02/2017). Research area: *Dynamic parallelism and elasticity*.

[2] **Chhaya Trehan** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (01/2015–05/2016). Research area: *Optimization algorithms for energy efficiency*.

[1] **Mahwish Arif** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast. Research area: *Performance portability*. (Co-supervised with Hans Vandierendonck.)

Visiting Scholars

[2] **Prof. Oscar Garcia Lorenzo** — Department of Computer Architecture, University of Santiago de Compostela (Apr. 2016). Topic: *Hardware counter-based performance analysis, modeling, and improvement through thread migration in NUMA systems*.

[1] **Dr. Satoshi Imamura** — System LSI Laboratory, Kyushu University. Research area: *Energy-efficiency optimization on multicore NUMA servers*.

M.Sc. Research Students

Primary Advisor

[29] **Katelyn Crumpacker** — Computer Science, Virginia Tech (May 2026). Thesis title: *LLM-guided run-time parameter optimization for energy-efficient model inference*.

[28] **Timothy Coyne** — Computer Science, Virginia Tech (May 2025). Thesis title: *Performance portability of CUDA across NVIDIA GPU architectures*.

[27] **Max Fisher** — Computer Science, Virginia Tech (May 2025). Thesis title: *Analysis of memory access patterns and prefetching strategies for large language model inference*.

[26] **Moustafa Kahla** — Electrical and Computer Engineering, Virginia Tech (Dec. 2023). Thesis title: *Automatic source code transformation to pass compiler optimization*.

[25] **Manthan Shah** — Electrical and Computer Engineering, Virginia Tech (Jun. 2023). Thesis title: *Analysis of YOLOv7 inference performance through a systems perspective and identification of optimization opportunities*.

[24] **Shreya Bhandare** — Computer Science, Virginia Tech (Jun. 2023). Thesis title: *Designing RDMA-based efficient communication for GPU remoting*.

[23] **Melissa Cameron** — Computer Science, Virginia Tech (Jun. 2023). Thesis title: *Parallel islands: a diversity-aware tool for parallel computing education*.

[22] **Ishaan Gulati** — Computer Science, Virginia Tech (Jun. 2023). Thesis title: *A scalable leader-based consensus algorithm*.

[21] **Matthew Jackson** — Computer Science, Virginia Tech (May 2023). Thesis title: *Computational offloading for real-time computer vision in unreliable multi-tenant edge systems*.

[20] **Daniel Moyer** — Computer Science, Virginia Tech (May 2021). Thesis title: *Punching holes in the cloud: direct communication between serverless functions using NAT traversal*.

[19] **Dimitris Chassapis** — Computer Science, University of Crete. Thesis title: *Static analysis for parallelism and correctness in task dataflow programming models*.

[18] **Ioannis Manousakis** — Computer Science, University of Crete (Jul. 2013). Thesis title: *TPROF: An energy profiler for task-parallel programs*.

[17] **Evangelos Kafentarakis** — Computer Science, University of Crete (Jul. 2013). Thesis title: *Lprof: A tool for profiling locality awareness in a task-based programming model*.

[16] **Christi Symeonidou** — Computer Science, University of Crete (Jul. 2013). Thesis title: *Distributed region-based allocation and synchronization*.

[15] **Kallia Chronaki** — Computer Science, University of Crete (Jun. 2013). Thesis title: *Exploiting pipelined parallelism with task dataflow programming models*.

[14] **Alexandros Labrineas** — Computer Science, University of Crete (Jun. 2013). Thesis title: *BDDT-SCC: A task-parallel runtime for the Single-Chip Cloud Computer*.

[13] **Anastasios Papagiannis** — Computer Science, University of Crete (Mar. 2013). Thesis title: *MapReduce on distributed-memory many-core architectures*.

[12] **Angelos Papatriantafyllou** — Computer Science, University of Crete (Mar. 2012). Thesis title: *Optimized block-based dependence analysis for task parallelism*.

[11] **Constantinos Koukos** — Computer Science, University of Crete (Aug. 2010). Thesis title: *Locality management in task-based parallel programming models*. (Co-supervised with Angelos Bilas.)

[10] **Pranav Tendulkar** — ALaRi Institute, Advanced Studies in Embedded Systems Design (Jun. 2010). Thesis title: *Runtime OpenMP support using hardware primitives on explicitly memory-managed multiprocessors*.

[9] **Michail Zampetakis** — Computer Science, University of Crete (Apr. 2010). Thesis title: *Runtime support for programming explicit-communication chip multiprocessors*.

[8] **Maria Katsamani** — Computer Science, University of Crete (Mar. 2010). Thesis title: *Software implementation of MPI primitives on multicore FPGA*. (Co-supervised with Manolis Katevenis.)

[7] **Benjamin Rose** — Computer Science, Virginia Tech (May 2009). Thesis title: *Intra- and inter-chip communication support for asymmetric multicore processors with explicitly managed memory hierarchies*.

[6] **Beran Nova Bryant** — Computer Science, Virginia Tech (May 2008). Thesis title: *Temperature-aware scheduling of parallel applications on shared-memory multiprocessors*.

[5] **Harshil Shah** — Computer Science, Virginia Tech (May 2008). Thesis title: *Application parallelization on the Cell/BE*.

[4] **Jyotirmaya Tripathi** — Computer Science, Virginia Tech (May 2008). Thesis title: *Scheduling parallel applications on paravirtualized shared-memory multiprocessors*.

[3] **Ankur Shah** — Computer Science, Virginia Tech (Apr. 2008). Thesis title: *Prediction models for multi-dimensional power-performance optimization on many cores*.

[2] **Scott Schneider** — Computer Science, College of William & Mary (Jun. 2005). Thesis title: *Factory: An object-oriented parallel programming substrate for deep multiprocessors*.

[1] **Robert McGregor** — Computer Science, College of William & Mary (May 2005). Thesis title: *Scheduling with bus bandwidth considerations on shared-memory multiprocessors*.

Co-Advisor

[4] **Foivos Zakkak** — Computer Science, University of Crete (Mar. 2012). Thesis title: *SCOOP: Language extensions and compiler optimizations for task-based programming models*. (Co-supervised with Angelos Bilas.)

[3] **Ioannis Kesapides** — Computer Science, University of Crete (Mar. 2011). Thesis title: *Dynamic dependence analysis on multi-core processors*. (Co-supervised with Angelos Bilas.)

[2] **Michail Alvanos** — Computer Science, University of Crete (Jun. 2010). Thesis title: *Design and evaluation of a task-based parallel H.264 video encoder for the Cell processor*. (Co-supervised with Angelos Bilas.)

[1] **George Tzenakis** — Computer Science, University of Crete (Oct. 2009). Thesis title: *Tagged procedure calls (TPC): efficient runtime support for task-based parallelism on the Cell processor*. (Co-supervised with Angelos Bilas.)

Undergraduate (BEng/BSc) Research Students

[43] **Kayden Moraes** — Computer Science, Virginia Tech (Fall 2025). Project: *Agentic AI for HPC code optimization*.

[42] **Adarsh Chittimoori** — Computer Science, Virginia Tech (Fall 2025). Project: *Agentic AI for HPC code optimization*.

[41] **Claire Shin** — Computer Science, Virginia Tech (Fall 2025). Project: *Agentic AI for HPC code optimization*.

[40] **Ariana Nafisi** — Computer Science, Virginia Tech (Fall 2025). Project: *Agentic AI for HPC code optimization*.

[39] **Krishnateja Reddy** — Computer Science, Virginia Tech (Spring 2025; Fall 2025). Project: *Agentic AI for HPC code optimization*.

[38] **Kathleen Reece** — Computer Science, Virginia Tech (Spring 2025; Fall 2025). Project: *Agentic AI for HPC code optimization*.

[37] **Bryan Torres** — Computer Science, Virginia Tech (Spring 2025; Fall 2025). Project: *Agentic AI for HPC code optimization*.

[36] **Veljko Cvetkovic** — Computer Science, Virginia Tech (Spring 2025; Fall 2025). Project: *Agentic AI for HPC code optimization*.

[35] **Yasir Hasan** — Computer Science, Virginia Tech (Fall 2024–Spring 2025). Project: *Agentic AI for HPC code optimization*.

[34] **Denise Cooney** — Computer Science, Virginia Tech (Spring 2025). Project: *Agentic AI for HPC code optimization*.

[33] **Asif Rahman** — Computer Science, Virginia Tech (Fall 2024–Spring 2025). Project: *Agentic AI for HPC code optimization*.

[32] **Aneesh Tummeti** — Computer Science, Virginia Tech (Fall 2024–Spring 2025). Project: *Agentic AI for HPC code optimization*.

[31] **Aidan Walters** — Computer Science, Virginia Tech (Fall 2024–Spring 2025). Project: *Agentic*

AI for HPC code optimization.

[30] **Jack Williamson** — Computer Science, Virginia Tech (Summer 2023; Fall 2023). Project: *Supporting flexible SIMD ISAs in WebAssembly*.

[29] **Anthony Nguyen** — Computer Science, Virginia Tech (Summer 2023; Fall 2023). Project: *WebAssembly performance profiling*.

[28] **Alex Lin** — Computer Science, Virginia Tech (Summer 2023; Fall 2023). Project: *Supporting GPUs in WebAssembly*.

[27] **Gianfranco Vivanco** — Computer Science, Virginia Tech (Summer 2023; Fall 2023). Project: *Performance characterization of LLM inference*.

[26] **Tatiana Monteiro** — Computer Science, Virginia Tech (Summer 2023). Project: *Optimization of inference from decision trees*.

[25] **Jamie Whiting** — Computer Science, Virginia Tech (Summer 2023). Project: *RISC-V simulation and system software*.

[24] **Riyos Pudasaini** — Computer Science, Virginia Tech (Summer 2023). Project: *Scalable memory allocators for Rust*.

[23] **Angel Gonzalez** — Computer Science, Virginia Tech (Summer 2023). Project: *Cache simulators*.

[22] **Kieran Siek** — Computer Science, Virginia Tech (Spring 2022). Project: *Code generation for matrix ISA extensions*.

[21] **Aditya Iyer** — Computer Science, Virginia Tech (Spring 2022). Project: *FaaS workload characterization*.

[20] **Parker Harnack** — Computer Science, Virginia Tech (Summer 2021). Project: *Performance analysis of persistent memory allocators*.

[19] **Lalitha Kupa** — Computer Science, Virginia Tech (Summer 2021). Project: *NUMA persistent memory management in the operating system*.

[18] **Ishaan Singh** — Computer Science, Virginia Tech (Summer 2020). Project: *Container caching*.

[17] **Nikolaos Parasyris** — Electrical and Computer Engineering, National Technical University of Athens (Sep. 2015). Project: *Fine-grain energy profiling of large software repositories*.

[16] **Stylianos Ninidakis** — Computer Science, University of Crete (Jun. 2011). Project: *Parallelizing irregular applications with task dataflow*.

[15] **Nikolaos Papakonstantinou** — Computer Science, University of Crete (Jun. 2011). Project: *Distributed dynamic dependence analysis for task dataflow models*.

[14] **Nikolaos Papadopoulos** — Computer Science, University of Crete (Feb. 2012). Project: *Scheduler-driven dynamic data placement for NUMA multi-cores*.

[13] **Ioannis Manousakis** — Computer Science, University of Crete (May 2011). Project: *Component-level power instrumentation on multiprocessors*.

[12] **Dimitrios Chassapis** — Computer Science, University of Crete (May 2011). Project: *Static dependence analysis for task dataflow models*.

[11] **Christi Symeonidou** — Computer Science, University of Crete (May 2011). Project: *Multi-node communication layer on the SARC FPGA prototype*.

[10] **Alexandros Labrineas** — Computer Science, University of Crete (May 2011). Project: *Early release optimizations for task dataflow programming models*.

[9] **Kallia Chronaki** — Computer Science, University of Crete (May 2011). Project: *Parallel loop scheduling on the SARC multi-core processor*.

[8] **Christos Margiolas** — Computer Science, University of Crete (Jun. 2010). Project: *Data placement and NUMA-aware optimization of MapReduce*.

[7] **Foivos Zakkak** — Computer Science, University of Crete (Jun. 2010). Project: *Source-to-source compiler optimizations for task parallelism*. (Co-supervised with Angelos Bilas.)

[6] **Spyros Tsatuhas** — Computer Science, University of Crete (Jun. 2010). Project: *POSIX threads library implementation on the SARC FPGA prototype*.
[5] **Evangelos Kafentarakis** — Computer Science, University of Crete (Jun. 2009). Project: *Software shared memory layer for CPU–GPU systems*.
[4] **Anastasios Papagiannis** — Computer Science, University of Crete (Jun. 2009). Project: *Performance analysis of virtual machine schedulers in Xen*.
[3] **Patrick Fiaux** — Computer Science, Virginia Tech (May 2007). Project: *Application parallelization and optimization on Cell/BE*.
[2] **James Dzierwa** — Computer Science, College of William & Mary (May 2006). Project: *Hardware monitors for power–performance adaptation*.
[1] **Evan McCreedy** — Computer Science, College of William & Mary (May 2004). Project: *Multilevel parallelization of MPIBlast*.

Service

Professional Activities

Membership in Professional Societies

- **IEEE Computer Society:** Fellow (2024–present); Distinguished Visitor (2024–present); Distinguished Contributor (2022–present); Senior Member (2011–2022); Member (1995–2011).
- **Association for Computing Machinery (ACM):** Distinguished Member (2018–present); Senior Member (2011–2018); Member (1995–2011).
- **Asia-Pacific Artificial Intelligence Association (AAIA):** Fellow (2024–present).
- **International Artificial Intelligence Industry Alliance (AIIA):** Fellow (2024–present).
- **British Computer Society (BCS):** Fellow (2014–present).
- **The Institution of Engineering and Technology (IET):** Fellow (2017–present).
- **United Kingdom Council of Professors and Heads of Computing (CPHC):** Member (2012–2019).
- **ACM SIGARCH:** Member (2000–present).
- **ACM SIGOPS:** Member (2000–present).
- **ACM SIGHPC:** Member (2000–present).
- **Chartered Engineer (CEng):** 2017–present.
- **Technical Chamber of Greece:** Member (1996–present).

Conference Committee Activities

Steering Committee Chair:
[2] ACM International Conference on Supercomputing (ICS) — 2024–present
[1] IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS) — 2022–2023

Steering Committee Associate Chair:
[1] ACM International Conference on Supercomputing (ICS) — 2023–2024

Program Chair/Co-Chair:

[9] ACM International Conference on High Performance Parallel and Distributed Computing (HPDC) — 2026
[8] ACM International Conference on Supercomputing (ICS) — 2023
[7] ACM International Conference on Supercomputing (ICS) — 2022
[6] International Workshop on Deployment and Use of Accelerators (DUAC) — 2021–2025
[5] IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGRID) — 2014
[4] International Workshop on Power-Aware Algorithms, Systems and Architectures (PASA) — 2013
[3] EuroMPI MPI Conference — 2011
[2] IEEE International Conference on Scalable Computing and Communications (ScalCom) — 2011
[1] Workshop on Parallel Programming for Accelerators (PPAC) — 2009–2011

Program Vice Chair/Area Chair:

[7] IEEE International Parallel and Distributed Processing Symposium (IPDPS) — Area Chair (Programming Models and Runtime Systems), 2026
[6] International European Conference on Parallel and Distributed Computing (EUROPAR) — Global Area Chair (Scheduling and Load Management), 2022
[5] International Conference on Embedded Computer Systems: Architecture, Modeling and Simulation (SAMOS) — 2016
[4] IEEE/ACM International Conference on High-Performance Computing, Networking, Architecture and Storage (SC) — 2014
[3] European Conference on Parallel Processing (EuroPar) — 2012
[2] IEEE International Parallel and Distributed Processing Symposium (IPDPS) — 2011
[1] International Conference on Parallel Processing (ICPP) — 2007

General Chair:

[10] IEEE International Conference on Cluster Computing (CLUSTER) — 2010; 2018
[9] IEEE International Symposium on the Performance Analysis of Systems and Software (ISPASS) — 2018
[8] ISC Conference Workshop on Approximate and Transprecision Computing on Emerging Technologies — 2018
[7] HiPEAC Conference Workshop on Approximate Computing — 2018
[6] SC Conference Workshop on Energy Efficient Supercomputing — 2013–2017
[5] ParCo Conference MiniSymposium on Edge Computing — 2017
[4] ParCo Conference MiniSymposium on Parallel Programming for Reliability and Energy Efficiency — 2016
[3] ParCo Conference MiniSymposium on Energy and Resilience in Parallel Programming — 2015
[2] HiPEAC Conference Workshop on Energy Efficient Heterogeneous Computing — 2015
[1] ICPP Conference Workshop on Power Aware Systems and Architectures — 2013

Program Committee:

[56] Annual Meeting of the Association for Computational Linguistics (ACL) — 2027
[55] IEEE/ACM International Symposium on High-Performance Computer Architecture (HPCA) — 2027
[54] IEEE/ACM International Conference on High-Performance Computing, Networking, Storage and Analysis (SC) — 2012–2018; 2026
[53] ACM International Conference on Architectural Support for Programming Languages and

Operating Systems (ASPLOS) — 2021; 2026
[52] Annual AAAI Conference on Artificial Intelligence (AAAI) — Senior Program Committee Member, 2026
[51] ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP) — 2013; 2015; 2020–2021; 2025–2026
[50] International Conference on Parallel Processing (ICPP) — 2003–2004; 2008; 2014; 2016–2021; 2024–2025
[49] IEEE/ACM International Symposium on Microarchitecture (MICRO) — 2024–2025 (External Review Committee)
[48] IEEE International Conference on Parallel Architectures and Compilation Techniques (PACT) — 2009; 2022–2023; 2025
[47] International Symposium on Computer Architecture (ISCA) — 2025
[46] IEEE/ACM International Symposium on High Performance Distributed Computing (HPDC) — 2019–2021; 2025
[45] ACM International Conference on Supercomputing (ICS) — 2007; 2009; 2011–2012; 2014; 2017; 2020–2021; 2025
[44] IEEE/SBC International Symposium on Computer Architecture and High-Performance Computing (SBAC-PAD) — 2025
[43] IEEE International Parallel and Distributed Processing Symposium (IPDPS) — 2013–2014; 2017–2021; 2023–2024
[42] European Conference on Parallel Processing (EuroPar) — 2013–2014; 2016; 2020; 2023–2024
[41] IEEE Big Data — 2013–2018; 2023–2024
[40] IEEE International Conference on Edge Computing (EDGE) — 2022
[39] IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGRID) — 2013; 2015–2016; 2018–2022
[38] International Supercomputing Conference (ISC) — 2018–2020
[37] International Green and Sustainable Computing Conference (IGSC) — 2019–2020
[36] Workshop on Languages and Compilers for Parallel Computing (LCPC) — 2020
[35] IEEE International Conference on High-Performance Computing (HiPC) — 2014–2016; 2019
[34] Parallel Computing Conference (ParCo) — 2015; 2017; 2019
[33] IEEE Graph Computing Conference (GC) — 2019
[32] ACM International Conference on Distributed Event-Based Systems (DEBS) — 2018
[31] International Conference on Embedded Computer Systems: Architecture, Modeling, and Simulation (SAMOS) — 2014–2015; 2017–2018
[30] ISC Conference Workshop on Communication Architectures for HPC, Big Data, Deep Learning, and Clouds at Extreme Scale (ExaComm) — 2017–2018
[29] ACM Conference on Computing Frontiers (CF) — 2011; 2014–2015; 2017
[28] IEEE Conference on Cluster Computing (CLUSTER) – 2016, 2015, 2013, 2012, 2011
[27] ISC Workshop on Energy-Aware High-Performance Computing (EnaHPC) – 2017, 2016, 2015, 2014
[26] IEEE International Conference on Big Data (BigData) – 2024, 2017, 2016, 2014
[25] IEEE International Conference on Ubiquitous Computing and Communications (IUCC) – 2016, 2012
[24] HiPEAC International Conference on High Performance and Embedded Architectures and Compilation – 2015, 2014, 2013, 2012
[23] ACM PPoPP Conference Workshop on General Purpose Processing using GPU (GPGPU) – 2015, 2014
[22] Feedback Computing Conference – 2015, 2010

[21] IEEE International Conference on Green Computing and Communications (GreenCom) – 2013, 2011, 2010
[20] IEEE International Conference on Parallel and Distributed Systems (ICPADS) – 2012, 2010, 2006, 2004
[19] IEEE High-Performance Computing and Communications Conference (HPCC) – 2012, 2009
[18] Symposium on Application Accelerators for High-Performance Computing (SAAHPC) – 2012, 2011, 2010, 2009
[17] IFIP Network and Parallel Computing Conference (NPC) – 2012, 2011, 2010
[16] International Conference on Architecture of Computer Systems (ARCS) – 2012, 2011, 2010
[15] IEEE Network and Storage Systems Conference (NAS) — 2009–2011
[14] IEEE International Conference on e-Business Engineering (ICEBE) — 2010–2011
[13] IEEE Green Computing Conference — 2010
[12] International Conference on Algorithms and Architectures for Parallel Processing (ICA3PP) — 2010
[11] IEEE International Conference on Scalable Computing and Communications (ScalCom) — 2009–2010
[10] IEEE Cloud Computing Conference (CloudCom) — 2009–2010
[9] IEEE International Conference on Computational Science and Engineering (CSE) — 2009–2010
[8] International Conference on Future Computational Technologies and Applications (Future Computing) — 2010
[7] International Conference on Parallel and Distributed Computing, Applications and Technologies (PDCAT) — 2010
[6] IEEE Autonomic and Trusted Computing Conference (ATC) — 2006–2008
[5] Balkan Conference on Informatics (BCI) — 2007
[4] ACM SIGMETRICS International Conference on Measurement and Modeling of Computer and Communication Systems — 2006
[3] IEEE International Conference on Pervasive Systems (ICPS) — 2005
[2] International Symposium on High Performance Computing (ISHPC) — 2003
[1] International Conference on Computational Science (ICCS) — 2001

Program Committee (Workshops):

[21] ICPP Conference Workshop on Heterogeneous and Unconventional Cluster Architectures and Applications (HUCAA) — 2015–2016
[20] ParCo Conference MiniSymposium on Parallel Programming for Reliability and Energy-Efficiency (PP4REE) — 2016
[19] HiPEAC Conference Workshop on Approximate Computing (WAPCO) — 2015–2016
[18] CGO Conference Workshop on Code Optimization for Multi- and Many-Cores (COSMIC) — 2014–2015
[17] HiPEAC Conference Workshop on Programmability and Architectures for Heterogeneous Multicores (MULTIPROG) — 2010–2015
[16] ICPP Conference Workshop on Parallel Programming Models and System Software for High-End Computing (P2S2) — 2013–2014
[15] SC Conference Energy-Efficient Supercomputing Workshop (E2SC) — 2013–2014
[14] ISC Conference Workshop on Virtualization for High-Performance Computing (VHPC) — 2008–2014
[13] PLDI Conference Workshop on Memory Systems Performance and Correctness (MSPC) — 2011; 2014
[12] IPDPS Workshop on Accelerators and Hybrid Exascale Systems (ASHES) — 2012–2014

[11] IBM Extreme Scale Parallel Architecture and Systems Workshop (ESPAS) — 2014
[10] Embedded Operating Systems Workshop (EWiLi) — 2013
[9] ISCA Conference Workshop on Future Architectural Support for Parallel Programming Workshop (FASPP) — 2011–2012
[8] HiPEAC Workshop on Computer Architecture and Operating System CoDesign (CAOS) — 2012
[7] PACT Conference Workshop on Programming Models for Emerging Architectures (PMEA) — 2009–2011
[6] IPDPS Conference Workshop on Software Engineering Innovation for HPC Clouds (SinHPC) — 2011
[5] ICS Conference Workshop on Characterizing Applications for Heterogeneous Exascale Systems (CACHES) — 2011
[4] International Workshop on Cloud Computing Interoperability and Services (InterCloud) — 2010
[3] ISCA Conference Workshop on Next Generation Multi/Many-core Technologies (FMT) — 2008; 2010
[2] IPDPS Conference High-Performance Power-Aware Computing Workshop (HPPAC) — 2008–2009
[1] IBM Cell Processor Programming Workshop (CELL) — 2009

Steering Committee Member:

[3] ACM International Conference on Supercomputing (ICS) — 2022–present
[2] IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS) — 2018–present
[1] IEEE International Conference on Cluster Computing (CLUSTER) — 2009–2011; 2017–2019

Other Committee Service:

[9] IEEE/ACM International Symposium on Microarchitecture (MICRO) — External Review Committee, 2024
[8] IEEE/ACM International Conference on High-Performance Computing, Networking, Storage and Analysis (SC) — Reproducibility Challenge Committee, 2021
[7] IEEE/ACM International Conference on High-Performance Computing, Networking, Storage and Analysis (SC) — HPC Impact Showcase Chair, 2017; Tutorials Committee, 2013–2014
[6] ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP) — External Review Committee, 2012; 2014
[5] International Supercomputing Conference (ISC) — Tutorials Committee, 2012–2014
[4] HiPEAC Conference — Workshop and Tutorials Chair, 2011
[3] EuroMPI Conference — Finance Chair, 2011
[2] ACM International Conference on Supercomputing (ICS) — Finance Chair, 2009; Workshops and Tutorials Chair, 2007
[1] SIAM Conference on Parallel Processing for Scientific Computing — Mini-Workshop Organizer, 2004; 2006

Session Chair

[15] ACM International Symposium on High-Performance Parallel and Distributed Computing (HPDC) — 2026
[14] ACM International Conference on Supercomputing (ICS) — 2020–2026 (Keynote sessions in 2022–2023)
[13] Workshop on Deployment and Use of Accelerators (DUAC) — 2022–2023
[12] IEEE/ACM International Symposium on High-Performance Distributed Computing (HPDC) — 2020–2021

[11] ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS) — 2020–2021
[10] IPDPS Workshop on Variability in Computing Systems (VarSys) — 2016
[9] SC Workshop on Energy-Efficient Supercomputing (E2SC) — 2014
[8] IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGRID) — 2014
[7] International Conference on Embedded Computer Systems: Architecture, Modeling and Simulation (SAMOS) — 2013
[6] ICPP Conference Workshop on Power Aware Systems and Architectures — 2013
[5] IEEE/ACM International Conference on High-Performance Computing, Networking, Analysis and Storage (SC) — 2012–2013
[4] ACM International Conference on Computing Frontiers (CF) — 2011
[3] International Conference on High Performance and Embedded Architectures and Compilation (HiPEAC) — 2008; 2011
[2] International Conference on Parallel Processing (ICPP) — 2010
[1] IEEE International Parallel and Distributed Processing Symposium (IPDPS) — 2009

Editorial Boards

[17] *Computer Physics Communications* — **HPC Specialist Editor**, 2017–present
[16] *Journal of Computational Science* — **Editorial Board Member**, 2014–present
[15] *International Journal of High Performance Computing Applications (IJHPCA)* — **Associate Editor**, 2012–present
[14] *Sustainable Computing: Informatics and Systems (SUSCOM)* — **Editorial Board Member**, 2010–present
[13] *Future Internet* — **Editorial Board Member**, 2020–2026
[12] *Frontiers in High-Performance Computing* — **Editor**, 2023–2024
[11] *IEEE Transactions on Parallel and Distributed Systems* — **Associate Editor**, 2018–2021
[10] *IEEE Transactions on Parallel and Distributed Systems* — **Best Paper Committee**, 2021
[9] *IEEE Transactions on Parallel and Distributed Systems* — **Artifact Evaluation Committee**, 2020–2021
[8] *International Journal of Parallel, Emergent and Distributed Systems (IJPEDS)* — **Associate Editor**, 2010–2019
[7] *International Journal of Information Technology, Communications and Convergence* — **Editorial Board Member**, 2009
[6] *Scientific Programming* — **Editorial Board Member**, 2015–2016
[5] *Concurrency and Computation: Practice and Experience (CCPE)* — **Editorial Review Board**, 2015
[4] *Parallel Computing (PARCO)* — **Guest Editor**, 2015
[3] *IET Computers and Digital Techniques* — **Guest Editor**, 2014
[2] *Sustainable Computing: Informatics and Systems (SUSCOM)* — **Guest Editor**, 2014
[1] *Journal of Autonomic and Trusted Computing* — **Editorial Board Member**, 2006–2007

Reviewer Work for Technical Journals, Conferences, and Publishers post 2020

[19] *IEEE Transactions on Parallel and Distributed Systems* (2026: 1 paper)
[18] *Future Generation Computer Systems* (2026: 1 paper)
[17] *ACM Transactions on Autonomous and Adaptive Systems* (2025: 1 paper)
[16] *IEEE Transactions on Services Computing* (2025: 1 paper)
[15] *Future Generation Computer Systems* (2025: 1 paper)
[14] *IEEE Transactions on Mobile Computing* (2025: 1 paper)

[13] *Frontiers in High-Performance Computing* (2024: 1 paper)
[12] *Future Generation Computer Systems* (2023: 1 paper)
[11] *Computer Physics Communications* (2022: 1 paper)
[10] *Engineering with Computers* (2022: 1 paper)
[9] *Software: Practice and Experience* (2022: 1 paper)
[8] *Concurrency and Computation: Practice and Experience* (2022: 1 paper)
[7] *International Journal of High-Performance Computing Applications* (2022: 1 paper)
[6] *Future Generation Computer Systems* (2021: 1 paper)
[5] *Journal of Parallel and Distributed Computing* (2021: 1 paper)
[4] *IEEE Access* (2021: 1 paper)
[3] *Bioinformatics* (2021: 1 paper)
[2] *IEEE Transactions on Parallel and Distributed Systems* (2020: 1 paper)
[1] *IEEE Transactions on Computers* (2020: 1 paper)

Reviewer Work for Technical Journals, Conferences, and Publishers pre 2020

[41] *ACM Transactions on Computer Systems*
[40] *ACM Transactions on Programming Languages and Systems*
[39] *ACM Transactions on Parallel Computing*
[38] *ACM Transactions on Architecture and Code Optimization*
[37] *ACM Transactions on Embedded Computing Systems*
[36] *ACM Transactions on Reconfigurable Technology and Systems*
[35] *ACM Computer Architecture Letters*
[34] *IEEE Micro*
[33] *IEEE Computer*
[32] *IEEE Spectrum*
[31] *IEEE Transactions on Parallel and Distributed Systems*
[30] *IEEE Transactions on Computers*
[29] *IEEE Access*
[28] *The Computer Journal*
[27] *Proceedings of the VLDB Endowment*
[26] *Journal of Parallel and Distributed Computing*
[25] *International Journal of Parallel Programming*
[24] *IBM Journal of Research and Development*
[23] *BMC Bioinformatics*
[22] *ETRI Journal*
[21] *IET Computers and Digital Techniques*
[20] *Journal of Signal Processing Systems*
[19] *International Journal of High-Performance Computing Applications*
[18] *Journal of Systems and Software*
[17] *Transactions on High Performance and Embedded Architectures and Compilation*
[16] *Journal of VLSI Signal Processing Systems*
[15] *Future Generation Computer Systems*
[14] *Scientific Programming*
[13] *EURASIP Journal on Embedded Systems*
[12] *Sustainable Computing: Informatics and Systems*
[11] *Software: Practice and Experience*
[10] *Journal of Network and Computer Applications*
[9] *Springer Nature* (reviewer)

[8] *Springer Computing* (reviewer)
[7] *Journal of Computer Science and Technology*
[6] *ACM Symposium on Parallel Algorithms and Architectures (SPAA)*
[5] *IEEE/ACM International Symposium on Microarchitecture (MICRO)*
[4] *IEEE/ACM International Symposium on High-Performance Computer Architecture (HPCA)*
[3] *IEEE International Symposium on Modeling, Analysis, and Simulation of Computer and Telecommunication Systems (MASCOTS)*
[2] *IEEE International Communications Conference (ICC)*
[1] *IEEE International Conference on Quantitative Evaluation of Systems (QEST)*

Significant University and Departmental Leadership and Service

[51] Computer Science Department, Qualifying Exam Committee Chair – Systems, Networking and Cybersecurity (2026)
[50] College of Engineering Promotion and Tenure Committee, Virginia Tech (Member 2025–present)
[49] Personnel Committee, Computer Science, Virginia Tech (Chair 2025–present)
[48] Computer Science Department Head Search Committee, Virginia Tech (2023–2024)
[47] Personnel Committee, Computer Science, Virginia Tech (Associate Chair 2024–present)
[46] Personnel Committee, Computer Science, Virginia Tech (Member 2022–present)
[45] University Proposal Development Institute (PDI) Mentor, Virginia Tech (2024–present)
[44] PI Eligibility Working Group, Virginia Tech (2022)
[43] Undergraduate Program Committee, Computer Science, Virginia Tech (2021–2022)
[42] Faculty Hiring Committee, Computer Science, Virginia Tech (2021)
[41] School of Computer Science Working Group, Virginia Tech (2020–2021)
[40] Operations Working Group, Computer Science, Virginia Tech (2020)
[39] University Reputation Group, Queen's University Belfast (2018)
[38] Global Challenges Research Fund Committee, Queen's University Belfast (2018)
[37] Director of the Queen's Global Research Institute on Electronics, Communications and Information Technology (ECIT) (2018–2019)
[36] Vice Chancellor Selection Panel, Queen's University Belfast (2017)
[35] Director of Information Services Selection Panel, Queen's University Belfast (2017)
[34] University Brand Committee, Queen's University Belfast (2017–present)
[33] Member of the University Research Forum, Queen's University Belfast (2012–2019)
[32] Chair of the University High-Performance Computing Advisory Group, Queen's University Belfast (2012–2019)
[31] Excellence in Leadership Training Program Speaker, Queen's University Belfast (2017)
[30] Member of the University Academic Council, Queen's University Belfast (2016–2018)
[29] Member of the Faculty of Engineering and Physical Sciences Promotions Panel, Queen's University Belfast (2016–2019)
[28] Member of the Faculty of Engineering and Physical Sciences Executive Board, Queen's University Belfast (2016–2019)
[27] Head of School of EEECS, Queen's University Belfast (2016–2018)
[26] Post-Doctoral Researchers Peer Mentor, Queen's University Belfast (2017–2018)
[25] Early-Career Academic Staff Peer Mentor, Queen's University Belfast (2012–2019)
[24] Senior Academic Staff Recruitment Working Group, Queen's University Belfast (2017–2019)
[23] Acting Director of Center for Data Science and Scalable Computing, Queen's University Belfast (2016–2019)
[22] Member of the Senior Management Group, Institute of Electronics, Communication and In-

formation Technologies, Queen's University Belfast (2016–2019)
[21] Engineering (Unit of Assessment 12) REF Champion, Queen's University Belfast (2018–2019)
[20] Computer Science (UoA11) REF Champion, Queen's University Belfast (2015–2019)
[19] School of EEECS Research Strategy Group Co-Chair, Queen's University Belfast (2015–2019)
[18] BCS Program Accreditation Committee Group Member, Queen's University Belfast (2013)
[17] Computer Science Building Project Implementation Group, Queen's University Belfast (2013–2017)
[16] Excellence in Leadership Training Program, Queen's University Belfast (2012–2013)
[15] Member of the Senior Management Group, School of EEECS, Queen's University Belfast (2012–2018)
[14] School of EEECS Education Committee, Queen's University Belfast (2012)
[13] School of EEECS Academic Staff Hiring Panels, Queen's University Belfast (2012–2019)
[12] Undergraduate Curriculum Committee in Computer Science, University of Crete (2010–2012)
[11] University Data Center Infrastructure Committee, University of Crete (2010–2012)
[10] Graduate Admissions Committee in Computer Science, University of Crete (2009–2011)
[9] Computer Organization Course Coordinator, Virginia Tech (2008–2009)
[8] Junior Faculty Mentor, Virginia Tech (2008–2009)
[7] Ph.D. Qualifying Exam Committee, Virginia Tech (2007–2009)
[6] Computer Science Computing Resources Committee, Virginia Tech (2006–2008)
[5] Computer Science Graduate Admissions Committee, College of William & Mary (2005–2006)
[4] Computer Science Graduate Curriculum Committee, College of William & Mary (2005–2006)
[3] Computer Science Faculty Hiring Committee, College of William & Mary (2002–2005)
[2] Computer Science Equipment Committee, College of William & Mary (2003–2005)
[1] Freshman Academic Advisor, College of William & Mary (2004–2006)

Significant External Leadership and Service

[45] Computing Research Association (CRA), Leadership Academy Committee (2024–present)
[44] IEEE Computer Society Senior Membership Reviewer (2025–2026)
[43] American Association for the Advancement of Science (AAAS), Session Reviewer (2024)
[42] External Accreditation and Evaluation Panel — Graduate Program (M.S. in Data Science and Machine Learning; M.S. in Information Systems), Hellenic Open University (2023)
[41] External Accreditation and Evaluation Panel — Undergraduate Program, Department of Informatics and Communications, University of Ioannina (2022)
[40] Scientific Advisory Board, Kevin Tier-2 HPC Center, Queen's University Belfast (2021–present)
[39] Faculty Appointment Committee, Norwegian University of Science and Technology (2022)
[38] External Evaluation Panel, Institute of Informatics and Telecommunications, National Center for Scientific Research "Demokritos" (2022)
[37] Evaluation Panel, Institute of Computer Science, Foundation for Research and Technology — Hellas (Seed Grants) (2021–present)
[36] Scientific Advisory Board, Marie Curie Individual Fellowship (2022)
[35] Faculty Appointment and Promotion Committee, University of Ioannina (2022)
[34] Faculty Appointment Committee, Heidelberg University (2021)
[33] External Accreditation and Evaluation Panel — Undergraduate Program, Department of Information Systems, Ionian University (2021)
[32] External Accreditation and Evaluation Panel — Undergraduate Program, School of Electrical and Computer Engineering, National Technical University of Athens (2021)
[31] Scientific Advisory Board, Institute of Computer Science, Foundation for Research and Technology Hellas (FORTH) (2020–present)

[30] Faculty Appointment and Promotion Committee, National Technical University of Athens (2020)
[29] Evaluation Panel, Foundation for Research and Technology — Hellas (Synergy Grants) (2019–present)
[28] Faculty Appointment and Promotion Committee, Technical University of Crete (2019)
[27] Faculty Appointment and Promotion Committee, University of Thessaly (2016; 2018–2019)
[26] Engineering and Physical Sciences Research Council (EPSRC) Strategic Advisory Team (SAT) Member — e-Infrastructure (2018–present)
[25] European Commission Framework Programme Consultant on Cloud Computing, European Commission DG Unit (2018)
[24] Faculty Appointment and Promotion Committee, Université Paris-Sorbonne (2018)
[23] Knowledge Transfer Partnership in Blockchain Technology, Vox Financial Partnerships (2018)
[22] Knowledge Transfer Partnership in OpenCL Technology, Crevinn (2017)
[21] Foreign Direct Investment Consultant, Invest Northern Ireland (2017)
[20] Faculty Appointment and Promotion Committee, Aristotle University of Thessaloniki (2017)
[19] Faculty Appointment and Promotion Committee, Chalmers University of Technology (2017)
[18] Faculty Appointment and Promotion Committee, University of Crete (2016–2018; 2020)
[17] Scientific Advisory Board, European Commission Horizon 2020 INTERTWinE Project (2015–2018)
[16] Scientific Advisory Member, European Commission Horizon 2020 Programme (2016)
[15] Grant Proposal Evaluator, King Abdullah University of Science and Technology (2016)
[14] Faculty Appointment and Promotion Committee, National Technical University of Athens (2014; 2016)
[13] Faculty Appointment and Promotion Committee, Ionian University (2016)
[12] Faculty Appointment and Promotion Committee, University of Athens (2015)
[11] Faculty Appointment and Promotion Committee, Technological Educational Institute of Athens (2015)
[10] Faculty Appointment and Promotion Committee, Technological Educational Institute of Piraeus (2014–2015)
[9] Faculty Appointment and Promotion Committee, Technological Educational Institute of Western Greece (2014–2016)
[8] Industry-Academic Partnership Training, Westminster Higher Education Forum (2013)
[7] External examiner, School of Computing, University of Leeds (2012–present)
[6] Faculty Appointment and Promotion Committee, University of Crete (2013)
[5] Multiple industrial consultancies, Queen's University Belfast (2012–present)
[4] Faculty Appointment and Promotion Committee, University of Ioannina (2011)
[3] Faculty Appointment and Promotion Committee, Aristotle University of Thessaloniki (2011)
[2] Faculty Hiring Committee, Technical University of Denmark (2010)
[1] Faculty Hiring Committee, National Technical University of Athens (2010)

Government Research Funding Panelist & Reviewer Service

[25] National Science Foundation (NSF) — Panelist, 2025 (two programs)
[24] Swiss National Science Foundation — Grant proposal reviewer, 2016; 2020; 2025
[23] French National Research Agency (ANR) — Panelist, 2025
[22] Hellenic Foundation for Research and Innovation (Greece) — Panelist, 2024–2025
[21] National Science Foundation (NSF), CISE — Panelist, 2021; 2023–2024
[20] Coastal Virginia Center for Cyber Innovation (COVA CCI, U.S.) — Grant proposal reviewer, 2021

[19] Knowledge Foundation (Sweden) — Grant proposal reviewer, 2021
[18] Italian National Research Council (CINECA) — Grant proposal reviewer, 2021
[17] European Research Council (ERC) — Grant proposal reviewer, 2020
[16] U.S. Department of Energy — Grant proposal reviewer, 2020
[15] Royal Academy of Engineering (U.K.) — Grant proposal reviewer, 2015; 2017
[14] Austrian Academy of Sciences — Grant proposal evaluator, 2017
[13] Technology Foundation STW (The Netherlands) — Grant proposal reviewer, 2016
[12] National Science Center (Poland) — Grant proposal reviewer, 2016
[11] Natural Sciences and Engineering Research Council of Canada (NSERC) — Discovery Grants panelist, 2014–2016
[10] University of Cyprus Research Foundation — Grant proposal reviewer, 2016
[9] Engineering and Physical Sciences Research Council (EPSRC, U.K.) — Grant proposal reviewer, 2012–2014; 2017–2018; 2021
[8] EPSRC (U.K.) — Platform Grant panelist, 2015
[7] European Commission FP7 Framework Programme — Project reviewer, 2013–2016
[6] European Commission FP7 Framework Programme — Grant proposal reviewer, 2012
[5] Greek Secretariat for Research and Technology — Grant proposal reviewer, 2010
[4] U.S.–Israel Binational Science Foundation — Grant proposal reviewer, 2009
[3] National Science Foundation (NSF) — Panelist, 2002–2004; 2008
[2] Natural Sciences and Engineering Research Council of Canada — Grant proposal reviewer, 2007
[1] Maryland Industrial Partnerships Program — Grant proposal reviewer, 2007

Conference Panels

[8] *Fifth International Workshop on Extreme Scale Programming Models and Middleware* (SC 2020) — Panelist: "When one size doesn't fit all: programming models in the post-Moore era" (Nov. 2020)
[7] *EPSRC Workshop on Manycore Computing: Hardware and Software* — Panelist (Southampton, U.K., Jan. 2018)
[6] *6th Workshop on Parallel Programming and Run-Time Management Techniques for Many-core Architectures (PARMA-DITAM 2015)* — Panelist (Jan. 2015)
[5] *IBM Research ExaChallenge Symposium* — Panelist (Dublin, Ireland, Oct. 2012)
[4] *39th International Conference on Parallel Processing (ICPP)* — Panelist: "Accelerators: fad, fashion, or future?" (Sep. 2010)
[3] *Fall Creek Falls Conference* — Panelist: "Key challenges presented by next generation hardware systems" (Sep. 2007)
[2] *Microsoft Faculty Research Summit* — Invitee (Redmond, WA, Sep. 2007)
[1] *15th ACM International Conference on Supercomputing (ICS)* — Panelist: NSF Next Generation Systems Software Program (Jun. 2001)

Public Engagement

[12] More to Know Podcast (Christopher Conover) — "The Future of Datacenters: AI, Energy Demand, and the Next Infrastructure Boom" (Feb. 2026). link
[11] TEDx MidAtlantic — "Myths and ways forward for AI" (Nov. 2025). link
[10] Northwest Indiana Times — "Data Centers in Northwest Indiana: Key Issues and Concerns" (Dec. 2025). link
[9] FuelCell Energy — "On the societal and economic implications of datacenters" (Dec. 2025). link

[8] MSN / WFXR — "On the societal and economic implications of datacenters" (Dec. 2025). link 1; link 2
[7] Cardinal News — "On the societal and economic implications of datacenters" (May 2025). link
[6] Roanoke Times (VT Science Corner) — "A Way to Democratize AI" (Jan. 2024). link
[5] Newswise — "Queen's Researchers in Bid to Develop the Fastest Supercomputers" (Feb. 2015). link
[4] NNTV Behind the Science — "On the importance of supercomputers" (2015)
[3] Horizon 2020 focus workshops in Northern Ireland and the U.K. (Mar.–Sep. 2014)
[2] CNN — "Faster than 50 Million Laptops: The Race Goes to Exascale" (Mar. 2012). link
[1] IEEE Computer (LEAV/Com) — "Big Iron Move Toward Exascale" (Oct. 2012). link

Invited Seminars and Talks

[55] *SWEET: A Hardware & Software Stack for Wrist-to-Cloud Intelligence* — NSF CSR Principal Investigators Meeting (Boca Raton, FL, Sep. 2025)
[54] *SWEET: System Architectures and Programming Abstractions for Resilient and Efficient Edge AI* — Discover-US Webinar, European Commission (virtual, Apr. 2025)
[53] *Sustainable AI from the Viewpoint of an AI Skeptic* — Distinguished Speaker Seminar, Department of Computer Engineering and Informatics (virtual, Dec. 2024)
[52] *Programming Models for High-Performance Computing: Key Drivers and Barriers to Adoption* — Invited talk, First NI-HPC Users Conference (virtual, Oct. 2021)
[51] *Why Should We Still be Concerned about Energy Management in System Software?* — Invited talk, 12th International Green and Sustainable Computing Conference (virtual, Oct. 2021)
[50] *From Approximate to Significance-Driven to Transprecision Computing: Challenges and Opportunities* — Distinguished Speaker Seminar, Department of Computational Science, ETH Zurich (Apr. 2018)
[49] *Virtualizing Any Accelerator Anywhere* — Distinguished Speaker Seminar, Department of Computer Science, College of William & Mary (Jan. 2018)
[48] *The Jevons Paradox in Computing Systems Research* — Distinguished Lecture Series, Department of Computer Science, Virginia Tech (Nov. 2016)
[47] *Computational Significance and its Implications for HPC* — 13th Workshop on Clusters, Clouds, and Data for Scientific Computing (CCDSC 2016) (Chemin de Chanzé, France, Oct. 2016)
[46] *Computational Significance and its Implications for Computing Systems* — School of Electrical and Electronic Engineering, Newcastle University (Oct. 2016)
[45] *Scaling Up, Out, or Down* — School of Informatics, University of Edinburgh (Mar. 2016)
[44] *Significance-Driven Runtime Systems* — RoMoL 2016 Workshop (Barcelona, Spain, Mar. 2016)
[43] *Advances in Energy-Efficient and Resilient HPC: Scaling Up, Out, or Back?* — Cardiff University (Mar. 2016)
[42] *Variability: Why should we care?* — Birds-of-a-Feather session on Variability in Large-Scale Computing Systems (SC 2015) (Austin, TX, Nov. 2015)
[41] *New Approaches to Energy-Efficient and Resilient HPC* — Department of Computer Science, Old Dominion University (Nov. 2015)
[40] *HPDC Research at Queen's: An Overview* — ARM High-Performance Computing Group (Manchester, U.K., Nov. 2015)
[39] *Server Resource Provisioning for Real-Time Analytics using Iso-Metrics* — Workshop on Performance Modeling: Methods and Applications (ISC 2015) (Frankfurt, Germany, Jul. 2015)
[38] *Evaluating Servers using Iso-Metrics: Power, Performance and Programmability Implications* — 8th Workshop on Programmability Issues for Heterogeneous Multicores (MULTIPROG 2015)

(Amsterdam, The Netherlands, Jan. 2015)

[37] *The Challenges and Opportunities of Micro-Servers in the HPC Ecosystem* — 12th Workshop on Clusters, Clouds, and Data for Scientific Computing (CCDSC 2014) (Chemin de Chanzé, France, Oct. 2014)

[36] *NVRAM as a User-Level Object Store* — HiPEAC Autumn Computing Systems Week (Athens, Greece, Oct. 2014)

[35] *On the Viability of Microservers for Real-Time Data Analytics* — HiPEAC Autumn Computing Systems Week (Athens, Greece, Oct. 2014)

[34] *NanoStreams: A Hardware and Software Stack for Real-Time Analytics on Fast Data Streams* — Horizon 2020: the HPC Opportunity (London, U.K., Mar. 2014)

[33] *GEMSCLAIM: Greener Mobile Systems by Cross-Layer Energy Management* — CHIST-ERA Projects Seminar (Istanbul, Turkey, Mar. 2014)

[32] *Searching for Data: The Ever-Increasing Role of Memory Hierarchies on the Performance and Sustainability of Computing Systems* — Inaugural lecture, Queen's University Belfast (Mar. 2013)

[31] *Energy as a Resource in Parallel Programs* — Supercomputing 2012 Birds-of-a-Feather: *Cool Supercomputing* (Nov. 2012)

[30] *Block-Level Dynamic Dependence Analysis for Task-Based Parallelism* — Workshop on Perspectives on Parallel Numerical Linear Algebra (Manchester, U.K., Jul. 2012)

[29] *Software Techniques for Energy Conservation in High-End Computing Systems* — Invited seminar, School of Computer Science, University of Manchester (U.K., Mar. 2012)

[28] *Energy Efficiency at Extreme Scale: Tools and Challenges* — Supercomputing 2011 Birds-of-a-Feather: *Energy-Efficiency* (Nov. 2011)

[27] *Rearchitecting MapReduce for Heterogeneous Multicore Processors with Explicitly Managed Memories* — School of Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Oct. 2010)

[26] *Determinism in Parallel Software and Architectures*. HiPEAC Systems Week Cluster Meetings, Barcelona, October 2009.

[25] *Parallelizing Non-trivial Applications with Multiple Programming Models* — HiPEAC Systems Week Cluster Meetings (Barcelona, Oct. 2009)

[24] *Uniform Evaluation of Programming Models* — HiPEAC Systems Week Cluster Meetings (Paris, Nov. 2008)

[23] *Unifying Layered Parallelism on the Cell BE* — Supercomputing 2007 Birds-of-a-Feather: *Unleashing the Power of the Cell Broadband Engine Processor for HPC* (Nov. 2007)

[22] *Unified Scheduling of Polymorphic Parallelism on Asymmetric Multi-core Systems* — Lawrence Livermore National Laboratory (Oct. 2007)

[21] *System Software for Scaling on Many Cores* — Oak Ridge National Laboratory (Sep. 2007)

[20] *Design and Implementation of Time- and Power-Efficient Software Stacks for Multicore Processors* — IBM Thomas J. Watson Research Center (Dec. 2006)

[19] *Design and Implementation of Time- and Power-Efficient Software Stacks for Multicore Processors* — Department of Computer Science, North Carolina State University (Sep. 2006)

[18] *Hardware Event-Driven Scalability Predictors: Improving Energy-Efficiency under Hard Performance Constraints on Multi-core and Multi-threaded Architectures* — Department of Electronic and Computer Engineering, Technical University of Crete (Jun. 2006)

[17] *Addressing the Challenges of Chip Multiprocessors using Autonomic Software* — Department of Computer Science, University of California, Riverside (Apr. 2006)

[16] *Addressing the Challenges of Chip Multiprocessors using Autonomic Software* — Department of Electrical and Computer Engineering, University of British Columbia (Mar. 2006)

[15] *High-Performance Power-Efficient Runtime Environments for Dense Computing Systems* — Depart-

ment of Computer Science, Virginia Tech (Feb. 2006)

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